NEET
SAMPLE PAPER 1 - 2020

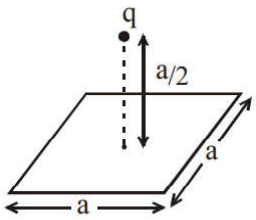
PHYSICS
SOLUTION

TRIPATHI STUDY ZONE
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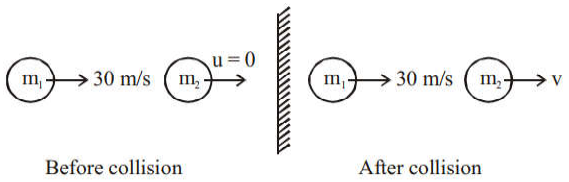
Answer Key

1.	(D)	2.	(B)	3.	(B)	4.	(B)	5.	(B)	6.	(B)	7.	(D)
8.	(A)	9.	(D)	10.	(B)	11.	(A)	12.	(B)	13.	(D)	14.	(A)
15.	(B)	16.	(C)	17.	(D)	18.	(A)	19.	(A)	20.	(C)	21.	(B)
22.	(D)	23.	(A)	24.	(A)	25.	(B)	26.	(D)	27.	(A)	28.	(A)
29.	(B)	30.	(A)	31.	(B)	32.	(A)	33.	(C)	34.	(D)	35.	(C)
36.	(A)	37.	(A)	38.	(C)	39.	(A)	40.	(B)	41.	(D)	42.	(A)
43.	(A)	44.	(A)	45.	(A)								

1. (D)



2. (B)



Since $m_1 >> m_2$, therefore heavy body (m_1) does not change its velocity after collision. By the definition of coefficient of restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 1 = \frac{v - 30}{30 - 0}$$

$\Rightarrow v = 60 m/s$

4. (B) $W = h\nu - \frac{1}{2}mv^2$

$h\nu$ = energy of incident photon

$$= \frac{12400}{1240} eV = 10 eV$$

$\therefore$ W = 10 – 8 = 2 eV

So, λ_0 = Threshold wavelength

$$= \frac{12400}{2eV} Å = 6200 Å.$$

5. (B) f = 5 cm, D = 25 cm

$$MP = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$$

6. (B) No of α particles (n) $\Rightarrow \frac{232 - 208}{4} = \frac{24}{4} = 6$

No of β particles :- $z - 2n(\alpha) + n'\beta = 82$

$90 - 2(6) + n'(\beta) = 82$

$n'(\beta) = 82 - 78 = 4$

7. (D) B, v and I should be mutually perpendicular.

9. (D) Velocity u of the body when it enters the liquid is given by

$$mgh = \frac{1}{2}mu^2 \quad \text{or} \quad u = \sqrt{2gh}$$

Let volume of the body = V

mass of the body = Vd

weight of the body = Vdg
Mass of liquid displaced = VD
Weight of liquid displaced = VDg
Net upward force = VDg–Vdg
= Vg(D–d)

$$\text{Retardation} = \frac{\text{net weight}}{\text{mass}}$$

$$= \frac{V(D-d)g}{Vd} = \left(\frac{D-d}{d}\right)g$$

$$\text{Acceleration } a = -\left(\frac{D-d}{d}\right)g$$

Final velocity, v in the liquid when the body is instantaneously at rest is zero. Let the time taken be t.

Now $v = u + at$

$$0 = \sqrt{2gh} - \left(\frac{D-d}{d}\right)gt \,,\ \left(\frac{D-d}{d}\right)gt = \sqrt{2gh}$$

$$\therefore\ t = \left[\frac{d}{D-d}\right]\sqrt{\frac{2h}{g}}$$

10. (B) Here, speed of sound, $v = 340\ ms^{-1}$
Length of pipe, $L = 17\ cm = 17 \times 10^{-2}\ m$
In a closed pipe (open at one end), the frequency of its nth harmonic is

$$v_n = \frac{nv}{4L} \quad \text{where } n = 1, 3, 5, 7, ...$$

Let n^{th} harmonic of closed pipe resonates with 1.5 kHz source.

$$\therefore\ v_n = 1.5 \times 10^3\ Hz$$

$$\therefore\ 1.5 \times 10^3 = \frac{nv}{4L}$$

$$n = \frac{1.5\times10^3\times4\times17\times10^{-2}}{340}$$

$$n = 3$$

11. (A) Component of $\vec{A}$ on $\vec{B}$ $= \dfrac{\vec{A}.\vec{B}}{|\vec{B}|} = \dfrac{2+3}{\sqrt{2}} = \dfrac{5}{\sqrt{2}}$

12. (B) Amplitude of electric field and magnetic field are related by the relation $\dfrac{E_0}{B_0} = c$

Average energy density of the magnetic field is

$$u_B = \frac{1}{4}\frac{B_0^2}{\mu_0}$$

$$= \frac{1}{4}\frac{E_0^2}{\mu_0 c^2} \qquad \left(\because B_0 = \frac{E_0}{c}\right)$$

$$= \frac{1}{4}\varepsilon_0 E_0^2 \qquad \left(\because c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}\right)$$

$$= \frac{1}{4} \times 8.854 \times 10^{-12} \times (2)^2$$

$$\approx 8.86 \times 10^{-12}\ \text{J m}^{-3}$$

13. (D) $C = \frac{\epsilon_0 A}{d}$; $C_1 = \frac{\epsilon_0 A}{2d} = \frac{C}{2}$

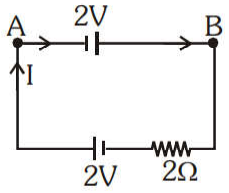
$$C_2 = \frac{5\,\epsilon_0 A}{2d} = \frac{5}{2}C$$

$$C_{eq} = C_1 + C_2 = 3C$$

$$\frac{\Delta C}{C}\% = \frac{2C}{C} \times 100 = 200\%$$

14. (A) $\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 3 & 5 \\ 3 & 4 & 6 \end{vmatrix} = -2\hat{i} + 45\hat{j} - 29\hat{k}$

15. (B) Circuit can be drawn as



Total emf = 2 + 2 = 4V

So $I = \frac{4}{2} = 2\text{A}$

17. (D) $P = 50\ \frac{N}{m^2} = 50 \times \frac{10^5}{(10^2)^2}\ \frac{\text{dyne}}{\text{cm}^2} = 500\ \frac{\text{dyne}}{\text{cm}^2}$

18. (A) Force on wire Q due to wire P is

$$F_P = 10^{-7} \times \frac{2 \times 30 \times 10}{0.1} \times 0.1 = 6 \times 10^{-5}\text{N}$$ (Towards left)

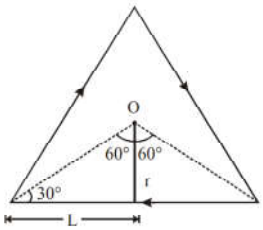
Force on wire Q due to wire R is

$$F_R = 10^{-7} \times \frac{2 \times 20 \times 10}{0.02} \times 0.1 = 20 \times 10^{-5}\text{N}$$ (Towards right)

Hence $F_{net} = FR - FP = 14 \times 10^{-5}\text{ N} = 1.4 \times 10^{-4}\text{ N}$ (Towards right)

19. (A) Magnetic field at O is

$$B = 3\left[\frac{\mu_0}{4\pi}\frac{i}{r}(\sin 60° + \sin 60°)\right]$$

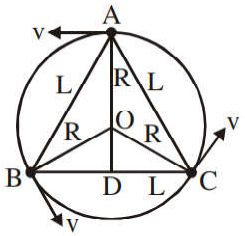


$$\tan 30° = \frac{r}{L} \qquad \Rightarrow r = \frac{L}{\sqrt{3}}$$

$$= 3\left(\frac{\mu_0}{4\pi}\right)\frac{(i)}{L}\left(\frac{\sqrt{3}}{2}\right)(\sqrt{3})\times 2$$

$$= \frac{9\mu_0 i}{4\pi L}$$

20. (C) Consider the circle with centre at O, and having radius R. In it consider three bodies of masses M each moving with velocity v under the action of their gravitational attraction



$$R = \frac{2}{3}AD = \frac{2}{3}\times L\sin 60° = \frac{2L}{3}\times\frac{\sqrt{3}}{2}$$

$$= \frac{L}{\sqrt{3}} \quad \text{or} \quad L = \sqrt{3}R$$

Centripetal force on any one mass M is

$$= \frac{2GM^2}{L^2}\cos\left(\frac{60°}{2}\right)$$

$$\therefore \quad \frac{Mv^2}{R} = \frac{2GM^2}{L^2}\times\frac{\sqrt{3}}{2}$$

$$\text{or} \quad \frac{GM^2}{(\sqrt{3}R)^2}\sqrt{3} = \frac{Mv^2}{R}$$

$$\therefore \quad v = \sqrt{\frac{GM}{\sqrt{3}R}}$$

22. (D) $\frac{\theta_1-\theta_2}{t}=-K\left(\frac{\theta_1+\theta_2}{2}-\theta_0\right)$

$t = 2 \text{ min} = 120 \text{ s}$

$\frac{80-70}{120}=-K\left(\frac{80+70}{2}-30\right)$ ...(i)

$\frac{60-50}{t}=-K\left(\frac{60+50}{2}-30\right)$ ...(ii)

(i)/(ii) $\Rightarrow$ (t = 216 s)

24. (A) Let v_1 and v_2 be the velocities of the body just before and just after the collision.

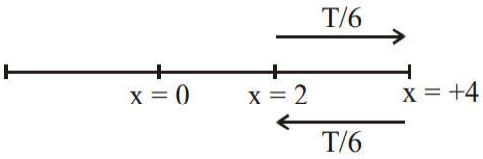
$KE_1 = \frac{1}{2}mv_1^2 = mgh_1$ .....(1)

$KE_2 = \frac{1}{2}mv_2^2 = mgh_2$ .....(2)

from (1) & (2) $\Rightarrow \frac{v_1^2}{v_2^2}=\frac{h_1}{h_2}=\frac{10}{2.5} = 4$

$\Rightarrow \frac{v_1}{v_2}=2$

25. (B)



minimum time taken by particle $= \frac{T}{6}+\frac{T}{6}$

$= \frac{T}{3}$

$= 0.4 \text{ s}$

26. (D) $P=\frac{V^2}{R}$ so, $R=\frac{V^2}{P}$ $\therefore R_1 = \frac{V^2}{100}$ & $R_2=R_3 = \frac{V^2}{60}$

Now, $P_1 = \frac{(250)^2}{(R_1+R_2)^2}\cdot R_1$ and

$P_2 = \frac{(250)^2}{(R_1+R_2)^2}\cdot R_2$ and $P_3 = \frac{(250)^2}{R_3}$

$P_1:P_2:P_3 = 15:25:64 \Rightarrow P_1 < P_2 < P_3$

27. (A) $dq = I\,dt = I_0\,(1 - e^{-t/T}) = \dfrac{E}{R}\,(1 - e^{-t/T})$

$$q = \int_0^T \frac{E}{R}(1-e^{-t/T})dt = \frac{E}{R}[t + Te^{-t/T}]_0^T$$

$$= \frac{E}{R}\left[T + \frac{T}{e} - T\right] = \frac{E}{R}\frac{T}{e}$$

28. (A) In Position-1

$$\tan\frac{\pi}{6} = \frac{\omega L}{R} = \frac{(1000)(\sqrt{3}\times 10^{-3})}{R} \text{ or } \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{R}$$

$\therefore \quad R = 3\Omega$

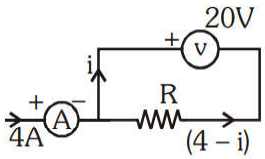
In Position-2

$$\tan\phi = \frac{X_C}{R} = \frac{(1/\omega C)}{R}$$

$$= \frac{1\Big/\left(1000\times\frac{1000}{3}\times 10^{-6}\right)}{3} = 1$$

$\therefore \quad \phi = \dfrac{\pi}{4}$

29. (B)



$(4 - i)\,R = 20$

$$R = \frac{20}{(4-i)}$$

for min. value of i $(R > 5)$

31. (B) $\beta = \dfrac{\lambda D}{d} \Rightarrow \beta \propto \dfrac{1}{d}$

32. (A) Revolutions per minute (N) $= \dfrac{3000}{\pi}$ rpm

so, frequency, $n = \dfrac{N}{60} = \dfrac{3000}{60\times\pi} = \dfrac{50}{\pi}$ Hz

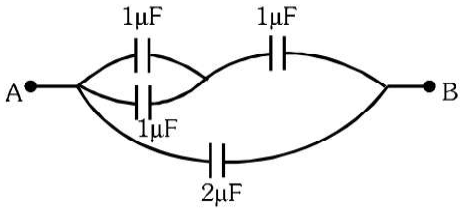
and $\omega = 2\pi n = 2\pi\times\dfrac{50}{\pi} = 100$ rad/sec

$I = 400$ kg m^2

so rotational kinetic energy (K) is

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}\times 400\times(100)^2 = 2\times 10^6 \text{ Joule}$$

33. (C)



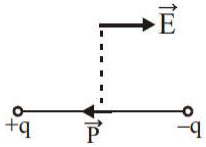
Redraw the ckt. $C_{AB} = \frac{8}{3}\mu F$

34. (D) $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \frac{36}{1}$

$\Rightarrow \frac{a_1 + a_2}{a_1 - a_2} = \frac{6}{1}$

$\Rightarrow \frac{a_1}{a_2} = \frac{7}{5}$

35. (C)



36. (A) $E = -\frac{dV}{dx} = -(10x + 10) = -(10 + 10)$

$= -20\ Vm^{-1}$

37. (A) Heat gain = Heat loose

$m_w s_w \Delta T = m_s L + m_w s_w (\Delta T)$

$40 \times 1 \times (54.3–25) = 2 \times L + 2 \times 1 \times (100–54.3)$

$L = 540.3$ cal/g

38. (C) $H_{max} = \frac{u^2 \sin^2\theta}{2g}$

According to problem

$\frac{u_1^2 \sin^2 45°}{2g} = \frac{u_2^2 \sin^2 60°}{2g}$

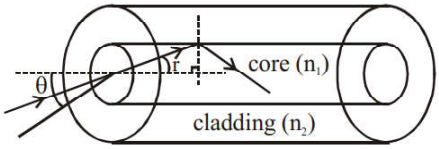
$\Rightarrow \frac{u_1^2}{u_2^2} = \frac{\sin^2 60°}{\sin^2 45°} \Rightarrow \frac{u_1}{u_2} = \frac{\sqrt{3}/2}{1/\sqrt{2}} = \sqrt{\frac{3}{2}}$

39. (A) The sixth coin is under the weight of four coins above it. Hence,

Reaction of the 6th coin on the 7th coin

= Force on the 6th coin due to 7th coin = 4mg

40. (B)



$1 \times \sin\theta = n_1 \times \sin r$

$$\sin r = \frac{\sin\theta}{n_1} \qquad ...(i)$$

$i > \theta_c$

$90 - r > \theta_c$

$\sin(90 - r) > \sin\theta_c$

$$\cos r > \frac{n_2}{n_1} \qquad ....(ii)$$

on squaring and then adding equ. (i) and eq. (ii), we get

$$\frac{\sin^2\theta}{n_1^2} + \frac{n_2^2}{n_1^2} \geq 1$$

$$\sin^2\theta + n_2^2 \geq n_1^2$$
$$\sin^2\theta \geq n_1^2 - n_2^2$$

$$\sin\theta \geq \sqrt{n_1^2 - n_2^2}$$

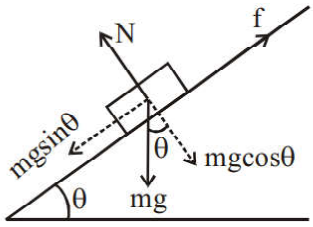
$$\theta \geq \sin^{-1}\sqrt{n_1^2 - n_2^2}$$

41. (D) $R = \dfrac{u^2 \sin 2\theta}{g} = \dfrac{2u_x v_y}{g}$

$\therefore$ Range $\propto$ horizontal initial velocity (u_x)

In path 4 range is maximum so football possess maximum horizontal velocity in this path.

42. (A) The various forces acting on the block are as shown in the figure.



From figure

$mg\sin\theta = f$ ....(i)

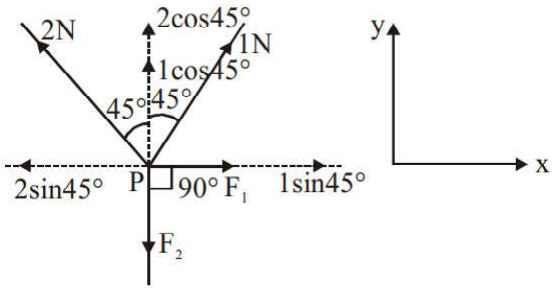
$mg\cos\theta = N$ ....(ii)

Divide (i) by (ii), we get

$$\tan\theta = \frac{f}{N} = \frac{\mu N}{N}$$

or $\theta = \tan^{-1}(\mu)$

43. (A)



Applying equilibrium conditions,

$\Sigma F_x = 0$

$\Rightarrow F_1 + 1\sin45° - 2\sin45° = 0$

or $F_1 = 2\sin45° - 1\sin45°$

$$= \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}\text{N}$$

and $\Sigma F_y = 0$

$\Rightarrow 1\cos45° + 2\sin45° - F_2 = 0$

$$F_2 = \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}}\text{N}$$

44. (A) Velocity of a wave is independent of others.